Bridging the Gap Between Theory and Practice in Adversarial Machine Learning: A Systematic Cross-Venue Analysis of 454 Papers (2022–2025)

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Abstract

Machine learning systems are increasingly used in security-critical contexts. As the field of adversarial machine learning (AML) has grown, questions have emerged about how well research aligns with the needs of practitioners. This systematic review covers 454 papers published between 2022 and 2025 at four leading security conferences (ACM CCS, IEEE S&P, NDSS and USENIX Security). We apply an adapted framework from Apruzzese et al. (2022) to evaluate how current work bridges the gap between theory and practice. We observe that a small but growing proportion of papers evaluate their methods on deployed systems (5.3%), while many studies still rely on assumptions that may not always hold in operational settings—for example, full gradient access, generous query budgets or white-box threat models. Identifying these assumptions highlights where future work can build stronger connections to real-world constraints. We identify five critical dimensions needing greater attention: clear threat modelling, accounting for limited attacker knowledge (no gradient access), validation on real systems, diversification beyond image classification, and consideration of economic and human factors. We also review notable real-world adversarial incidents and discuss emerging challenges in large language and multi-modal models. We close by offering constructive recommendations for researchers, conference organisers, industry practitioners and funding agencies to help align adversarial ML research with operational realities.

Keywords: adversarial machine learning, theory–practice gap, security research, systematic review, threat modelling

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1 Introduction

Machine learning (ML) is ubiquitous in modern software systems, powering applications from facial recognition and financial fraud detection to autonomous vehicles and content moderation. The advent of large language models has expanded ML's reach into areas such as customer service, code generation and medical decision support. As adoption has grown, so too has interest in their security and robustness. Researchers have shown that carefully crafted inputs—known as adversarial examples—can cause ML systems to behave in unexpected ways. The field of adversarial machine learning (AML) explores these vulnerabilities and develops strategies to mitigate them.

Over the past decade, researchers have demonstrated a wide range of attacks (e.g. imperceptible perturbations that cause misclassification, poisoned training data that implants backdoors, and privacy attacks that extract sensitive information) and proposed numerous defences. However, a recurring question in both academia and industry is how well these studies reflect the realities of deployed ML systems. Practical deployments often involve complex pipelines, limited attacker access, economic constraints and human factors that are not easily captured in purely theoretical work.

In 2022, Apruzzese et al. [1] drew attention to this mismatch by noting that real attackers rarely possess the full gradient information assumed in many studies, and that simple, low-tech methods often suffice in practice. Our work builds on this observation by systematically analysing AML papers published between 2022 and 2025 in four top security venues. We aim to quantify the gap between theory and practice and to highlight positive examples where research is moving closer to deployment realities. Throughout this paper we frame our findings as opportunities for growth: rather than criticising prior work, we focus on how the community can build on existing insights to make ML systems more secure in the real world.

The remainder of this paper is structured as follows. Section **??** summarises foundational concepts in adversarial ML, focusing on common attack types, defences and threat models. Section **??** describes our data collection and coding methodology. Sections **??** and **??** present quantitative and qualitative findings, respectively. Section **??** discusses emerging work on large language and multi-modal models. Section **??** explores temporal trends over the review period. We then offer recommendations and reflections on limitations before concluding.

2 Background: Foundations of Adversarial Machine Learning

Early research into adversarial ML revealed that deep learning models can be surprisingly sensitive to small, targeted changes in input data. Szegedy et al. [2] and Goodfellow et al. [3] showed that tiny perturbations, often imperceptible to humans, could cause state-of-the-art image classifiers to make high-confidence mistakes. These findings motivated a rich literature exploring why such vulnerabilities arise and how to defend against them.

A key insight from this early work is that adversarial robustness can be studied using optimisation. Attackers often formulate the problem of crafting an adversarial example as maximising a model's error subject to constraints on the perturbation magnitude. Popular norms such as L_0, L_2 and L_∞ measure different notions of perturbation size: L_0 counts how many features are changed, L_2 measures Euclidean distance and L_∞ bounds the maximum change to any single feature. Attack algorithms such as the Fast Gradient Sign Method (FGSM) [3], DeepFool [5] and Carlini–Wagner (C&W) attacks [6] use

gradient-based optimisation to find perturbations that satisfy these norms. Defences such as adversarial training [9] incorporate adversarial examples into the training process, while certified methods [13] aim to provide formal guarantees of robustness within specified threat models.

Over time, the AML community has broadened its scope beyond simple evasion attacks. Researchers have studied training-time threats such as poisoning and backdoors [19, 29], privacy attacks that attempt to extract information about the training data [16], and physical-world attacks that survive sensor noise and environmental transformations [23, 24]. The threat landscape also encompasses black-box and grey-box adversaries, who may only be able to query a model rather than inspect its internals. Recognising these diverse threats has spurred development of gradient-free and low-query attacks [36, 40] and defences suited to more realistic deployment scenarios.

A crucial element in any security study is the threat model—the attacker's knowledge, capabilities and objectives. White-box models grant full access to the model parameters and gradients, while black-box models permit only input–output queries. Real-world attackers often occupy a middle ground: they may know a model's architecture but lack its weights, or they may be limited by query costs and detection mechanisms. Clear threat modelling helps contextualise results and ensures that defences are evaluated under conditions that reflect their intended use.

This brief overview highlights how the field has matured, moving from proof-of-concept attacks in controlled settings towards more varied threat models and robust defences. Yet, as our analysis in the following sections shows, there remain opportunities to align research more closely with the complexities of operational systems.

3 Methodology

We conducted a systematic review of AML papers published between 2022 and 2025 in four premier security conferences: ACM CCS, IEEE Symposium on Security and Privacy (S&P), NDSS and USENIX Security. Together these venues contribute a comprehensive snapshot of security-focused AML research. Our dataset comprises 454 papers: 118 from CCS, 79 from S&P, 49 from NDSS and 208 from USENIX Security.

To assemble this dataset, we searched conference proceedings and digital libraries using keywords (e.g. "adversarial," "backdoor," "poisoning," "robustness") and cross-referenced prior reviews. We manually verified each candidate to ensure its primary focus was adversarial ML. For each paper we coded several attributes, drawing on the taxonomy of Apruzzese et al. [1] and adding practical considerations. Key coding dimensions included: the research focus (attack or defence), attack type, data domain, threat model (white-box, black-box or grey-box), reliance on gradient access, assumed query budget, use of real system testing, release of code and discussion of economic or organisational factors. A simple "Gap Score" aggregated six yes/no factors indicating how far a paper's assumptions diverged from typical deployment constraints (e.g. requiring gradient access or a high query budget). Scores range from 0 (close to practice) to 6 (largely theoretical).

While coding necessarily simplifies complex contributions, it enables quantitative trends to be discerned. We supplemented this analysis with qualitative reading to identify themes and positive examples that indicate where the field is progressing.

4 Quantitative Findings

Our quantitative analysis provides insight into how current AML research aligns with practical concerns. Figure 2 summarises five key indicators: real-world testing, gradient reliance, threat model assumptions, query budgets and code release.

4.1 Real-World Evaluation

Only **5.3%** of papers (24 out of 454) included any experiments on a deployed system or production-like environment. While this figure indicates that real-world testing remains rare, it also highlights promising examples where researchers have partnered with industry or adapted their methods to operational settings. Such work demonstrates the feasibility and value of real-world validation. As more organisations recognise the benefits of adversarial robustness, we hope to see this proportion grow.

4.2 Reliance on Gradients

Two-thirds of papers (**67.8%**) relied on gradient information from the target model. Many classic attacks and certified defences are formulated in terms of gradients, which makes them powerful in worst-case analyses. At the same time, gradient-free techniques are gaining traction, reflecting the recognition that real attackers may not have white-box access. Gradient-independent toolkits and transformation-based methods [36] offer an encouraging direction for future research.

4.3 Threat Model Assumptions

White-box threat models predominated (**63.2%** of papers), with black-box scenarios appearing in roughly one-third of papers and grey-box settings being comparatively rare. These figures suggest an opportunity: expanding evaluation to include limited knowledge adversaries will make defences more applicable to real deployments. Some recent work has begun to explore this space, for example by modelling partial knowledge or restricted query budgets, which we view as a positive development.

4.4 Query Budgets and Efficiency

Among studies that involved querying a target model, most assumed generous budgets, with **80.4%** falling into our high-query category. While many theoretical contributions are developed in this setting, practical attacks must often operate under strict rate limits and cost constraints. Emerging low-budget attacks such as HARDBEAT [40] and BounceAttack [41] demonstrate that efficiency and effectiveness are not mutually exclusive.

4.5 Data Domains and Reproducibility

Approximately **65%** of papers focused on image-based tasks, with text, audio, malware and other domains making up the remainder. This concentration reflects the historical roots of AML in computer vision but suggests opportunities to address underexplored modalities. Encouragingly, nearly **90%** of papers released code or artefacts, facilitating reproducibility and extension. Releasing code is an important step towards transparency and practical uptake.

4.6 Gap Score Overview

The average Gap Score across all papers was **3.17** (on a 0–6 scale). Most papers scored in the mid-range, combining some practical considerations with idealised assumptions. Only about 10% of papers scored 0 or 1 (very close to deployment conditions). This distribution underscores the mixed nature of the field: theory and practice continue to inform each other, but the majority of work could benefit from closer alignment with real-world constraints.

5 Qualitative and Thematic Findings

Our qualitative review surfaced several recurring themes. First, there is often a trade-off between robustness and utility. Many defences reduce model accuracy on clean data or increase computational overhead. Recent work strives to minimise this cost, for example through patch-based cleansing [43] or mitigations that maintain classification performance while reducing vulnerability [48].

Second, evaluation methodology matters. Metrics based solely on L_p norms do not fully capture whether perturbations are noticeable to humans or whether they have real-world consequences. Studies incorporating human perception (e.g. in autonomous driving simulators [50]) or domain-specific cost functions (e.g. in fraud detection [42]) offer a richer picture of robustness.

Third, system integration poses unique challenges. Attacks and defences that succeed in isolation may behave differently when incorporated into larger pipelines with pre- and post-processing, rate limiting and human oversight. Some of the most impactful adversarial incidents in practice exploit system-level interactions rather than pixel-level perturbations. For example, adversarial audio that evades YouTube's copyright system [35] demonstrates the need to consider platform-specific processing.

Fourth, real-world adversarial incidents often involve social engineering or system misconfiguration rather than sophisticated optimisation. Recent cases—such as hidden prompts in documents causing large language model assistants to leak information [60] or supply-chain attacks on coding assistants [59]—underscore that the weakest link may lie in how ML models are integrated with external data sources and tools.

Finally, cross-disciplinary collaboration is increasingly important. Addressing economic and organisational factors, understanding human perception, and designing effective guardrails for large language models all require input from beyond core ML security. Several papers have started to explore these dimensions [16, 27], and we view this broader engagement as a positive sign of maturation.

6 Foundation Models and Large Language Models

The rapid adoption of large language models (LLMs) and other foundation models introduces new dimensions to adversarial ML. These models operate on discrete tokens, are often accessed via APIs, and are integrated into complex workflows such as coding assistants and autonomous agents. As a result, traditional perturbation-based attacks are complemented by prompt engineering and cross-modal exploits.

Early research on LLM robustness often assumed white-box access to gradients, enabling optimisation in token space. More recent work recognises that commercial APIs restrict such access and emphasises black-box strategies. Automated techniques such as prompt-pairing and tree-of-thought

methods [67, 68] require only a handful of queries to generate effective jailbreaks. Multi-modal attacks that hide malicious instructions in images or audio [69, 81] demonstrate how vulnerabilities can cross channels.

Defences are evolving in response. In addition to prompt filtering and output moderation, secondary models (e.g. LlamaGuard [86]) can judge the safety of responses. Constitutional AI [91] leverages a model's own feedback to correct itself. Automated red-teaming frameworks [89] help discover failure cases at scale. While the LLM security landscape is still young, these efforts illustrate how theory and practice can inform each other: by testing new guardrails on deployed systems, researchers uncover practical challenges and refine threat models.

7 Temporal Trends

Our dataset spans four years (2022–2025), allowing us to examine changes over time. Although real-world testing remained relatively rare throughout, the proportion of studies considering low-query or black-box settings increased slightly. The appearance of LLM-focused security work in 2024–2025 diversified the modalities under study. Nonetheless, the core distribution of assumptions (e.g. reliance on gradients and white-box models) changed gradually rather than dramatically. These trends suggest steady but incremental progress towards greater practical relevance.

8 Recommendations

Our findings point to concrete steps that different stakeholders can take to bridge the theory–practice gap:

Researchers. When proposing new attacks or defences, explicitly discuss the threat model and justify assumptions about attacker knowledge and resources. Where feasible, evaluate methods under realistic constraints such as limited queries, partial knowledge and human perception. Partnering with practitioners or using publicly available platforms for testing can provide valuable insight into deployment challenges. Sharing code, data and evaluation scripts aids reproducibility and encourages adoption.

Conference organisers and reviewers. Encourage submissions to include some form of practical evaluation or discussion of applicability. Clear threat modelling should be part of the review criteria. Special tracks or awards recognising impactful real-world evaluations could incentivise researchers to go beyond benchmark datasets.

Practitioners. Document and share adversarial incidents or unusual inputs encountered in production systems, where confidentiality permits. Collaborations with academic researchers—such as bug bounties or red teaming—can help refine assumptions and test proposed defences. When assessing a new technique, consider its resource requirements, integration complexity and how it fits within organisational processes.

Funding agencies. Support interdisciplinary projects that bring together ML security experts with human–computer interaction researchers, economists and legal scholars. Funding calls could prioritise work that includes evaluation on real systems or that addresses underexplored modalities such as tabular or graph data. Shared benchmarks that approximate production conditions (e.g. API access, cost constraints, diverse data) would provide useful testbeds for future research.

These recommendations are intended as constructive guidance. By aligning incentives and sharing insights across communities, we believe the field can continue to advance both theoretical understanding and practical impact.

9 Limitations

This study has limitations. Our analysis focused on security conferences; ML conferences and domain-specific venues may reveal additional patterns. Coding complex papers into binary categories inevitably oversimplifies nuanced contributions. Our Gap Score treats all factors equally, whereas some may be more important than others depending on context. Because our dataset ends in mid-2025, newer developments are not captured. Finally, practicality is context-dependent: what is infeasible today may become routine as technology and attacker capabilities evolve. We encourage readers to interpret our findings as indicative rather than definitive and to build upon them with further analysis.

10 Conclusion

Our systematic review of 454 adversarial ML papers reveals both progress and opportunities for growth. While many studies remain rooted in idealised assumptions, a growing number engage with real-world constraints, release code and explore diverse modalities. We have highlighted key dimensions—threat modelling, limited attacker access, real-world evaluation, domain diversification, and economic or human factors—where increased attention could substantially improve the practical relevance of future work. By recognising the value of both theoretical insight and deployment-oriented research, we believe the AML community can accelerate the development of ML systems that are robust against adversarial misuse. Securing ML is not solely a technical problem but a socio-technical one; meeting this challenge will require collaboration across disciplines and between academia and industry.

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A Complete Paper Analysis Dataset

For completeness we provide the coded dataset used in our analysis. The table below summarises key attributes for each of the 454 papers. The binary fields indicate whether a paper satisfied a given criterion (✓) or not (×). Threat models and other categorical labels are abbreviated for compactness.

Table 1: Summary of all reviewed papers with key taxonomy fields. (G1: Focus; G2: Attack Type; G3: ML Type; G4: Domain; G5: Economics; G6: Code; G7: Real-world eval; T1: Threat Model; T2: Training Data Access; Q1: Gradient; Q2: Query Budget; Q3: Computation; WB: White-box flag; Gap Score.)

Year	Venue	First Author	G1	G2	G3	G4	G5	G6	G7	T1	T2	Q1	Q2	Q3	WB	Gap
2022	ACM	Using	def	Po	DL	Img	✓	✓	×	G	F	✓	H	L	0	3
2022	ACM	Reverse-Engineering	atk	Ev	DL	Aud	✓	✓	×	B		×	H	L	0	2
2022	ACM	Tianshuo	def	Pr	DL	Img	✓	✓	×	B		×	L	L	0	1
2022	ACM	CISPA	atk	Pr	DL	Oth	✓	×	×	B		×	L	L	0	2
2022	ACM	CISPA	atk	Pr	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2022	ACM	School	atk	Ev	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2022	ACM	Physical	atk	Ev	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2022	ACM	Framework	def	Ev	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2022	ACM	Florian	atk	Po	DL	Img	✓	×	×	W	F	✓	H	L	1	5
2022	ACM	via	atk	Pr	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2022	ACM	ExamplesforLeakage	def	Ev	DL	Img	✓	✓	×	W	P	✓	H	L	1	4
2022	ACM	Harnessing	atk	Ev	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2022	ACM	Feature	atk	Pr	DL	Oth	✓	×	✓	B		×	L	L	0	1
2022	ACM	Imperial	atk	Pr	Both	Oth	✓	✓	×	B		×	H	L	0	2
2022	ACM	Learning	atk	Ev	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2022	ACM	CISPA	atk	Ev	DL	Txt	✓	✓	×	B		×	H	L	0	2
2022	ACM	Amrita	def	Po	DL	Img	✓	×	×	B		✓	H	L	0	4
2022	ACM	A	def	Ev	Both	Txt	✓	✓	×	B		×	L	L	0	1
2022	ACM	CISPA	atk	Pr	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2022	ACM	ETH	def	Pr	Both	Oth	✓	×	×	B		×	L	L	0	2
2022	ACM	Hanqing	atk	Ev	DL	Aud	✓	✓	×	W	F	✓	H	L	1	4
2022	ACM	Stevens	atk	Pr	DL	Oth	✓	✓	×	W	F	✓	H	L	1	4
2022	ACM	Are	atk	Pr	DL	Oth	✓	✓	×	W	F	✓	H	L	1	4
2022	ACM	When	atk	Ev	DL	Aud	✓	✓	✓	B		×	H	H	0	1
2022	ACM	Learning	atk	Pr	DL	Img	✓	✓	×	B		×	H	L	0	2
2022	ACM	Using	atk	Po	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2022	ACM	Neural	atk	Ev	DL	Txt	✓	✓	×	B	P	✓	H	H	0	3
2022	ACM	CISPA	atk	Pr	DL	Img	✓	✓	×	B	P	×	L	L	0	1
2022	ACM	A	atk	Pr	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2022	ACM	LPGNet:	def	Pr	DL	Oth	✓	✓	×	B		✓	H	L	0	3
2022	IEEE	Encoders	atk	Po	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2022	IEEE	1Illinois	atk	Ev	DL	Img	✓	✓	×	B		×	L	L	0	1
2022	IEEE	Graph	atk	Pr	DL	Oth	✓	✓	×	B	F	✓	H	L	0	3
2022	IEEE	Imperceptible	atk	Ev	DL	Txt	✓	✓	✓	B		×	L	L	0	0
2022	IEEE	Security	def		DL	Oth	✓	✓	×	B		×		L	0	1
2022	IEEE	Jialuo	def		DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2022	IEEE	Yun	atk	Ev	DL	Oth	✓	✓	×	B		×	H	L	0	2
2022	IEEE	Production	atk	Po	DL	Img	✓	×	×	W	P	✓	H	L	1	5
2022	IEEE	Classification	def		Both	Mal	✓	✓	×	B		×	L	L	0	1
2022	IEEE	in	def	Po	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2022	IEEE	Markus	def	Pr	Both	Oth	✓	✓	×	B		×	L	L	0	1
2022	IEEE	Eugene	atk	Po	DL	Txt	✓	✓	×	B		✓	H	H	0	3
2022	IEEE	"Adversarial	atk	Ev	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2022	IEEE	Attacks	atk	Pr	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2022	IEEE	Weight	atk	Pr	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2022	IEEE	SoK:	def		Both	Oth	✓	×	×	B		×		L	0	2
2022	IEEE	Borja	atk	Pr	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2022	IEEE	Transformer	atk	Ev	DL	Txt	✓	✓	×	W	F	✓	H	L	1	4
2022	IEEE	Property	atk	Po	DL	Img	✓	✓	×	B	F	×	H	L	0	2
2022	IEEE	Yuhao	atk	Ev	DL	Img	✓	✓	×	B		×	H	L	0	2
2022	NDS	School	atk	Ev	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2022	NDS	Federated	def	Po	DL	Txt	✓	×	×	W		×	L	L	1	3
2022	NDS	Property	atk	Pr	DL	Img	✓	✓	×	B	F	✓	H	L	0	3
2022	NDS	RamBoAttack:	atk	Ev	DL	Img	✓	✓	×	B		×	H	H	0	2
2022	NDS	Shengwei	atk	Pr	DL	Img	✓	✓	✓	W	F	✓	H	L	1	3
2022	NDS	Robustness	both	Po	DL	Img	✓	×	×	W	F	✓	H	L	1	5
2022	NDS	Ahmed	atk	Po	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2022	USENIX	for	def	Ev	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2022	USENIX	Vocal	def	Ev	DL	Aud	✓	✓	×	B		×		L	0	1
2022	USENIX	31st	atk	Pr	Both	Oth	✓	×	×	W	F	✓	H	L	1	5
2022	USENIX	August	atk	Ev	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2022	USENIX	sponsored	atk	Po	DL	Img	✓	✓	×	W	F	✓	H	L	1	4

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Year	Venue	First Author	G1	G2	G3	G4	G5	G6	G7	T1	T2	Q1	Q2	Q3	WB	Gap
2022	USENIX	August	atk	Pr	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2022	USENIX	sponsored	atk	Pr	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2022	USENIX	sponsored	atk	Po	DL	Txt	✓	✓	×	W	F	✓	H	L	1	4
2022	USENIX	Attacks	atk	Po	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2022	USENIX	sponsored	def	Ev	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2022	USENIX	August	atk	Pr	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2022	USENIX	August	atk	Pr	DL	Oth	✓	✓	×	W	F	✓	H	L	1	4
2022	USENIX	Novel	atk	Pr	Both	Oth	✓	✓	×	B		×	H	L	0	2
2022	USENIX	Technische	def		Both	Mal	✓	×	×	B		×		L	0	2
2022	USENIX	1CISPA	atk	Pr	DL	Oth	✓	✓	×	W	F	✓	H	L	1	4
2023	ACM	CISPA	def	Ev	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2023	ACM	Machine	def	Pr	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2023	ACM	Virginia	atk	Po	DL	Img	✓	✓	×	G	P	✓	L	L	0	2
2023	ACM	through	atk	Pr	DL	Aud	✓	✓	×	W	F	✓	H	L	1	4
2023	ACM	Stealing	atk	Ev	DL	Txt	✓	✓	×	B		×	H	L	0	2
2023	ACM	Stolen	both	M	DL	Oth	✓	✓	×	B	P	✓	H	L	0	3
2023	ACM	Information	atk	Ev	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2023	ACM	Deep	def	Ev	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2023	ACM	Unforgeability	def		DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2023	ACM	Stefano	def	Ev	Traditional	Img	✓	✓	×	B		×	L	L	0	1
2023	ACM	Learning	atk	Po	DL	Oth	✓	✓	×	G	P	✓	H	L	0	3
2023	ACM	Secure	atk	Ev	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2023	ACM	Detection	atk	Ev	DL	Mal	✓	✓	×	B		×	L	L	0	1
2023	ACM	Code	def		DL	Oth	✓	✓	×	G		×	L	L	0	1
2023	ACM	and	def	Po	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2023	ACM	through	atk	Pr	DL	Oth	✓	✓	×	B		✓	H	L	0	3
2023	ACM	Jingxuan	def		DL	Oth	✓	✓	×	B		×	L	L	0	1
2023	ACM	and	atk	Ev	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2023	ACM	Evading	atk	Ev	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2023	ACM	Changqing	atk	Ev	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2023	ACM	AntiFake:	def	Ev	DL	Aud	✓	×	✓	B		×	H	L	0	2
2023	ACM	against	def	Po	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2023	IEEE	Effects	def	Po	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2023	IEEE	Deep	def	Ev	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2023	IEEE	arXiv:2112.04558v2	def	Ev	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2023	IEEE	Federated	atk	Ev	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2023	IEEE	SNAP:	atk	Po	DL	Oth	✓	✓	×	B		×	H	L	0	2
2023	IEEE	Federated	def	Po	DL	Img	✓	×	×	W	F	✓	H	L	1	5
2023	IEEE	Information	atk	Pr	DL	Txt	✓	✓	×	B		×	H	L	0	2
2023	IEEE	Covert	atk	Po	DL	Img	✓	✓	×	B		✓	H	L	0	3
2023	IEEE	DepthFake	atk	Ev	DL	Img	✓	✓	×	B		×	H	L	0	2
2023	IEEE	Jiameng	atk	Ev	DL	Txt	✓	✓	×	B		×		L	0	1
2023	IEEE	Multi-party	def	Pr	Both	Img	✓	✓	×	W	F	✓	H	L	1	4
2023	IEEE	Andrew	def	Pr	Both	Oth	✓	×	×	W	F	✓	H	L	1	5
2023	IEEE	Information	atk	Pr	DL	Txt	✓	✓	×	B		×	H	L	0	2
2023	IEEE	Extraction	atk	Ev	DL	Img	✓	✓	×	B		×	H	L	0	2
2023	IEEE	Limin	atk	Po	DL	Mal	✓	✓	×	W	F	✓	H	L	1	4
2023	IEEE	Learning	def	Po	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2023	IEEE	Attacks	both	Ev	DL	Txt	✓	✓	×	W	F	✓	H	L	1	4
2023	IEEE	Automatic	atk	Ev	DL	Aud	✓	✓	×	W	F	✓	H	L	1	4
2023	IEEE	yCybersecurity	def		DL	Img	✓	×	×	W		×	L	L	1	3
2023	IEEE	Ruijie	atk	Ev	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2023	IEEE	A	both	Pr	DL	Img	✓	✓	×	B	P	×	H	L	0	2
2023	IEEE	SNAP:	atk	Po	DL	Oth	✓	×	×	B		×	H	L	0	3
2023	IEEE	Hong	def	Ev	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2023	IEEE	BayBFed	def	Po	DL	Img	✓	×	×	W	F	✓	H	L	1	5
2023	IEEE	Andre	atk	Ev	DL	Aud	✓	✓	×	B	F	×	L	L	0	1
2023	IEEE	KASTEL	atk	Ev	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2023	IEEE	ELSA:	def	Po	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2023	IEEE	Shengwei	atk	Ev	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2023	IEEE	Patch-agnostic	def	Ev	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2023	IEEE	RAB:	def	Po	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2023	IEEE	ETH	def	Po	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2023	IEEE	and	def	Pr	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2023	IEEE	through	atk	Ev	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2023	IEEE	StyleFool:	atk	Ev	DL	Img	✓	✓	×	B		×	L	L	0	1
2023	NDS	DNN	def	Ev	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2023	NDS	Hadi	def	Ev	DL	Aud	✓	✓	×	B		×	H	L	0	2

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Year	Venue	First Author	G1	G2	G3	G4	G5	G6	G7	T1	T2	Q1	Q2	Q3	WB	Gap
2023	NDS	Yugeng	atk	Po	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2023	NDS	Commercial	atk	Ev	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2023	NDS	Classifiers	def	Ev	DL	Img	✓	✓	×	B		×	L	L	0	1
2023	NDS	Learning	def	Ev	DL	Oth	✓	✓	×	B		✓	H	L	0	3
2023	NDS	Jiayun	atk	Ev	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2023	NDS	Klim	atk	Ev	DL	Oth	✓	×	×	B		×	H	H	0	3
2023	NDS	Siyuan	atk	Po	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2023	NDS	Universidad	def	Po	DL	Txt	✓	✓	×	W	F	✓	H	L	1	4
2023	NDS	Deep	atk	Ev	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2023	NDS	Chunyi	atk	Pr	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2023	NDS	Robust	def	Po	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2023	NDS	Physical	def	Ev	Both	Oth	✓	✓	✓	G	P	×	L	L	0	0
2023	NDS	*Technische	def	Pr	Both	Txt	✓	✓	×	W	F	✓	H	L	1	4
2023	USENIX		atk	Ev	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2023	USENIX	Detection	atk	Ev	DL	Mal	✓	×	×	B		×	H	L	0	3
2023	USENIX	Scientific	atk	Pr	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2023	USENIX	of	def		DL	Oth	✓	✓	×	B		×	L	L	0	1
2023	USENIX	for	atk	Po	Both	Oth	✓	×	×	W	F	✓	H	L	1	5
2023	USENIX	via	def	Pr	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2023	USENIX	FREE	def	Ev	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2023	USENIX	1CISPA	def	Ev	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2023	USENIX	Jonathan	atk	Ev	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2023	USENIX	Vehicles	atk	Ev	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2023	USENIX	Fairness	atk	Ev	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2023	USENIX	Fairness	atk	Ev	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2023	USENIX	Decompiling	atk	Ev	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2023	USENIX	August	def		Both	Mal	✓	✓	×	B		×	L	L	0	1
2023	USENIX	Mazal	def	Ev	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2023	USENIX	August	def		DL	Mal	✓	✓	×	W	F	✓	H	L	1	4
2023	USENIX	is	atk	Pr	DL	Img	✓	✓	×	B	P	✓	H	L	0	3
2023	USENIX	is	def	Ev	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2023	USENIX	Federated	def	Po	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2023	USENIX	Kunal	atk	Ev	DL	Mal	✓	✓	×	B	P	×	H	L	0	2
2023	USENIX	for	def	Po	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2023	USENIX	by	def	Ev	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2023	USENIX	is	atk	Ev	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2023	USENIX	is	atk	Ev	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2023	USENIX	August	atk	Po	DL	Txt	✓	×	×	B		×	H	L	0	3
2023	USENIX	Universal	atk	Ev	DL	Img	✓	✓	×	B		×	H	L	0	2
2023	USENIX	is	atk	Ev	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2023	USENIX	August	def	Po	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2024	ACM	from	atk	Ev	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2024	ACM	Xueluan	def	Pr	DL	Img	✓	×	×	B		×	L	L	0	2
2024	ACM	SafeEar:	def	Pr	DL	Aud	✓	✓	×	B		×	L	L	0	1
2024	ACM	Against	atk	Ev	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2024	ACM	Multi-Agent	atk	Ev	DL	Img	✓	✓	×	B		×	H	H	0	2
2024	ACM	Text-to-Image	def	Ev	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2024	ACM	Autonomous	def	Ev	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2024	ACM	Wei	atk	Po	Both	Oth	✓	✓	×	G	P	✓	H	L	0	3
2024	ACM	Byzantine-Robust	def	Po	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2024	ACM	Recognition	atk	Ev	DL	Aud	✓	✓	✓	B		×		H	0	0
2024	ACM	Zijin	def	Ev	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2024	ACM	CSIRO's	atk	Po	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2024	ACM	Items	atk	Po	DL	Oth	✓	✓	×	W	F	✓	H	L	1	4
2024	ACM	Applications	atk	Pr	DL	Txt	✓	✓	×	B		✓	H	H	0	3
2024	ACM	Attack	atk	Ev	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2024	ACM	Attachment	atk	Ev	DL	Mal	✓	✓	✓	B		×	H	L	0	1
2024	ACM	Information	atk	Ev	DL	Txt	✓	✓	×	W	F	✓	H	L	1	4
2024	ACM	Alchemy:	def	Ev	DL	Img	✓	✓	×	W		✓	H	L	1	4
2024	ACM	S2NeRF:	both	Pr	DL	Img	✓	✓	×	B		✓	H	L	0	3
2024	ACM	Certified	atk	Po	DL	Oth	✓	✓	×	W	F	✓	H	L	1	4
2024	ACM	ETH	atk	Pr	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2024	ACM	Jie	atk	Pr	DL	Img	✓	✓	×	B	P	×	H	L	0	2
2024	ACM	Understanding	atk	Po	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2024	ACM	in	def	Ev	DL	Mal	✓	✓	×	W	F	✓	H	L	1	4
2024	ACM	A	def	Pr	DL	Img	✓	✓	×	B		×	H	L	0	2

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Year	Venue	First Author	G1	G2	G3	G4	G5	G6	G7	T1	T2	Q1	Q2	Q3	WB	Gap
2024	ACM	The	def	Ev	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2024	ACM	Fisher	def	Po	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2024	ACM	into	atk	Pr	DL	Txt	✓	✓	×	B		×	H	L	0	2
2024	ACM	BadMerging:	atk	Po	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2024	ACM	Optimization-based	atk	Ev	DL	Txt	✓	✓	×	W	F	✓	H	L	1	4
2024	ACM	Optimization-based	atk	Ev	DL	Txt	✓	✓	×	W	F	✓	H	L	1	4
2024	ACM	Model	atk	Pr	DL	Txt	✓	✓	×	W	F	✓	H	L	1	4
2024	ACM	Shenzhen	atk	Pr	DL	Txt	✓	✓	×	W	F	✓	H	L	1	4
2024	ACM	Information	def	Ev	DL	Img	✓	✓	×	B		×		H	0	1
2024	ACM	Learning	atk	Pr	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2024	ACM	Blind	def		Both	Aud	×	×	×	B		×		L	0	3
2024	ACM	Information	atk	Pr	DL	Txt	✓	×	×	B		×	L	L	0	2
2024	ACM	Deepfake	def		DL	Aud	✓	✓	×	B		×		L	0	1
2024	ACM	Membership	atk	Pr	DL	Img	✓	✓	×	B	P	×	L	L	0	1
2024	ACM	Membership	atk	Pr	DL	Img	✓	✓	×	B	P	×	L	L	0	1
2024	ACM	Neural	atk	Pr	DL	Img	✓	✓	×	B		×		L	0	1
2024	ACM		atk	Ev	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2024	ACM	Fine-grained	def	Ev	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2024	ACM	from	atk	Ev	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2024	ACM	Institute	atk	Pr	DL	Img	✓	✓	×	B		×	H	L	0	2
2024	ACM	Institute	atk	Pr	DL	Img	✓	✓	×	B	F	×	H	L	0	2
2024	ACM	of	def	Po	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2024	ACM	School	atk	Ev	DL	Txt	✓	✓	×	W	F	✓	H	L	1	4
2024	ACM	Examples:	atk	Ev	DL	Img	✓	✓	×	B		×	H	L	0	2
2024	ACM	Models	atk	Ev	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2024	ACM	Learning	atk	Po	DL	Img	✓	✓	×	B	P	×	L	L	0	1
2024	IEEE	Pattern	def	Po	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2024	IEEE	Robin	def	Pr	Both	Oth	✓	✓	×	W	F	✓	H	L	1	4
2024	IEEE	Yuzheng	def	Pr	Both	Img	✓	✓	×	B		×	L	L	0	1
2024	IEEE	Joshua	atk	Pr	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2024	IEEE	Ziqi	def	Pr	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2024	IEEE	Linguistic	atk	Ev	DL	Aud	✓	✓	✓	B		×	L	L	0	0
2024	IEEE	Language	atk	Ev	DL	Txt	✓	✓	×	W	F	✓	H	L	1	4
2024	IEEE	It's	def	Ev	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2024	IEEE	into	atk	Ev	DL	Img	✓	✓	×	B		×	H	L	0	2
2024	IEEE	Test-Time	atk	Po	DL	Img	✓	✓	×	G	P	✓	L	L	0	2
2024	IEEE	Xingshuo	atk	Po	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2024	IEEE	Mohammad	atk	Po	DL	Img	✓	×	×	G	P	✓	H	L	0	4
2024	IEEE	Attacks	def	Ev	DL	Img	✓	✓	×	B		×	H	L	0	2
2024	IEEE	Distribution	atk	Po	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2024	IEEE	Dropout	atk	Po	DL	Img	✓	×	×	W	F	✓	H	L	1	5
2024	IEEE	GROV	def	Ev	DL	Oth	✓	✓	×	B		×	L	L	0	1
2024	IEEE	Mahmoud	atk	Ev	DL	Oth	✓	✓	×	B	P	×	L	L	0	1
2024	IEEE	Alec	def	Ev	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2024	IEEE	Accelerator	atk	Pr	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2024	IEEE	KASTEL	atk	Ev	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2024	IEEE	Adversarial	def	Ev	DL	Txt	×	✓	×	W		×		L	1	3
2024	IEEE	2Services	atk	Ev	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2024	NDS	Large	def	Ev	DL	Txt	✓	✓	×	W	F	✓	H	L	1	4
2024	NDS	DEMASQ:	def	Ev	DL	Txt	✓	×	×	B		✓	H	L	0	4
2024	NDS	Attacks	atk	Pr	DL	Aud	✓	✓	×	W	F	✓	H	L	1	4
2024	NDS	Poisoning	atk	Po	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2024	NDS	by	def	Pr	DL	Img	✓	✓	×	B		×	L	L	0	1
2024	NDS	CamPro	atk	Ev	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2024	NDS	Ensemble	atk	Ev	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2024	NDS	Gelei	atk	Ev	DL	Txt	✓	×	×	B		×	H	H	0	3
2024	NDS	Watermarking	def	Ev	DL	Aud	✓	✓	✓	W	F	×	L	L	1	1
2024	NDS	ICS	atk	Ev	DL	Oth	✓	✓	×	W	F	✓	H	L	1	4
2024	NDS	Federated	def	Po	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2024	NDS	Chengkun	def	Po	DL	Txt	✓	✓	×	B		✓	L	L	0	2
2024	NDS	Recognition	atk	Ev	DL	Aud	✓	✓	×	W	F	✓	H	L	1	4
2024	NDS	Transpose	atk	Pr	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2024	NDS	Transpose	atk	Pr	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2024	NDS	Inversion-based	def	Pr	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2024	NDS	Networks	atk	Po	DL	Img	✓	✓	×	W	F	✓	H	L	1	4

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Year	Venue	First Author	G1	G2	G3	G4	G5	G6	G7	T1	T2	Q1	Q2	Q3	WB	Gap
2024	NDS	Browser	def	Pr	Traditional	Oth	✓	×	×	B		×		L	0	2
2024	NDS	Deep	def	Ev	DL	Img	✓	✓	×	B		×	H	L	0	2
2024	NDS	Bang	def	Pr	DL	Oth	✓	✓	×	B		×	L	L	0	1
2024	NDS	Against	atk	Po	DL	Img	✓	✓	×	B	P	✓	H	L	0	3
2024	NDS	Speaker	atk	Ev	DL	Aud	✓	✓	✓	B		×		L	0	0
2024	NDS	Mitigating	def	Po	DL	Img	✓	×	×	B		×		L	0	2
2024	NDS	for	def		DL	Mal	✓	✓	×	B		×	L	L	0	1
2024	NDS	in	both	Ev	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2024	NDS	Attacks	def	Po	DL	Txt	✓	✓	×	W	F	×	L	L	1	2
2024	NDS	Chaoxiang	atk	Ev	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2024	USENIX	Tong	atk	Ev	DL	Txt	✓	✓	×	B		×	L	L	0	1
2024	USENIX	Attacks	atk	Ev	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2024	USENIX	Approach	def	Po	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2024	USENIX	Qingzhao	atk	Ev	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2024	USENIX	Zhenghang	def		Both	Oth	✓	✓	×	B		×		L	0	1
2024	USENIX	Synthesis	def	Pr	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2024	USENIX	Measuring	def		DL	Aud	✓	✓	×	B		×	L	L	0	1
2024	USENIX	Large	atk	Ev	DL	Txt	✓	✓	×	B		×	H	L	0	2
2024	USENIX	Efficiency	def	Ev	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2024	USENIX	Efficiency	def	Ev	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2024	USENIX	Kathrin	atk	Ev	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2024	USENIX	DNN-GP:	both	Ev	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2024	USENIX	for	atk	Ev	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2024	USENIX	DeepEclipse	atk	Ev	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2024	USENIX	DeepEclipse	atk	Ev	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2024	USENIX	Between	atk	M	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2024	USENIX	Red-Teaming	atk	Ev	DL	Img	✓	✓	×	B		×	H	L	0	2
2024	USENIX	Red-Teaming	atk	Ev	DL	Img	✓	✓	×	B		×	H	L	0	2
2024	USENIX	Property	atk	Pr	DL	Img	✓	✓	×	B	F	×	H	L	0	2
2024	USENIX	Transposed	def		DL	Img	✓	×	×	W	F	✓	H	L	1	5
2024	USENIX	in	def	Pr	DL	Img	✓	✓	×	B		×	L	L	0	1
2024	USENIX	Shuaifan	def	Pr	DL	Img	✓	✓	✓	W	F	✓	H	L	1	3
2024	USENIX	CISPA	atk	Ev	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2024	USENIX	CISPA	atk	Ev	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2024	USENIX	Runtime	atk	Ev	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2024	USENIX	Splitting	def	Ev	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2024	USENIX	Yixin	atk	Pr	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2024	USENIX	Changjiang	atk	Po	DL	Img	✓	✓	×	B	F	✓	H	L	0	3
2024	USENIX	Guangsheng	atk	Pr	DL	Img	✓	×	×	W	F	✓	H	L	1	5
2024	USENIX	Bailey	def	Pr	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2024	USENIX	Meng	atk	Ev	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2024	USENIX	"Minxue	def	Pr	DL	Img	✓	✓	×	W		✓	H	L	1	4
2024	USENIX	CISPA	atk	Pr	DL	Img	✓	✓	×	B	P	×	H	H	0	2
2024	USENIX	False	atk	Ev	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2024	USENIX	Shenchen	atk	Ev	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2024	USENIX	through	atk	Pr	DL	Oth	✓	×	×	B		✓	H	L	0	4
2024	USENIX	Shaofeng	atk	Ev	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2024	USENIX	Mudjacking:	def	Po	DL	Img	✓	×	×	B		✓	H	L	0	4
2024	USENIX	Randomised	atk	Ev	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2024	USENIX	33rd	atk	Ev	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2024	USENIX	is	def		DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2024	USENIX	August	def	Pr	Both	Txt	✓	✓	×	B		×	L	L	0	1
2024	USENIX	August	atk	Pr	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2024	USENIX	is	atk	Po	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2024	USENIX	is	def		DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2024	USENIX	is	def	Po	DL	Img	✓	×	×	B		×		L	0	2
2024	USENIX	Attacks	def	Pr	DL	Img	✓	✓	×	B		✓	H	L	0	3
2024	USENIX	Model	atk	Ev	DL	Txt	✓	✓	×	W	F	✓	H	L	1	4
2024	USENIX	Zooming	both	Ev	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2024	USENIX	August	atk	Pr	DL	Txt	✓	✓	×	B		×	L	L	0	1
2024	USENIX	is	def		DL	Txt	✓	✓	×	W	F	✓	H	L	1	4
2024	USENIX	August	def		DL	Oth	✓	✓	×	W	F	×	H	L	1	3
2024	USENIX	August	def		DL	Oth	✓	✓	×	W	F	✓	H	L	1	4
2024	USENIX	August	def	Ev	DL	Img	✓	✓	×	B		×		L	0	1
2024	USENIX	August	def		Both	Mal	✓	×	×	B		×	L	L	0	2
2024	USENIX	Learning	def	Ev	Both	Oth	✓	✓	×	W	F	✓	H	L	1	4
2024	USENIX	Framework	def		DL	Txt	✓	✓	×	B		×		L	0	1
2024	USENIX		def		DL	Img	✓	✓	×	B		×		L	0	1

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Year	Venue	First Author	G1	G2	G3	G4	G5	G6	G7	T1	T2	Q1	Q2	Q3	WB	Gap
2024	USENIX		atk	Ev	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2024	USENIX	Inference	def	Pr	DL	Img	✓	×	×	G		✓	H	L	0	4
2025	ACM	1Technion	atk	Ev	DL	Txt	✓	✓	×	B		×	H	L	0	2
2025	ACM	in	atk	Ev	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2025	ACM	arXiv:2501.05928v2	atk	Po	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2025	ACM	Anti-Facial	atk	Ev	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2025	ACM	BIFOLD	atk	Ev	DL	Oth	✓	✓	×	W	F	✓	H	H	1	4
2025	ACM	Targeted	atk	Ev	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2025	ACM	Language	atk	Po	DL	Txt	✓	✓	×	B		×	L	L	0	1
2025	ACM	Models	def	Ev	DL	Img	✓	×	✓	G	P	✓	H	H	0	3
2025	ACM	Adversarial	atk	Po	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2025	ACM	Construction	atk	Ev	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2025	ACM	Latent-based	atk	Ev	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2025	ACM	Milad	atk	Ev	DL	Mal	✓	✓	✓	W	F	✓	H	L	1	3
2025	ACM	VillainNet:	atk	Po	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2025	ACM	Deep	def		DL	Mal	✓	✓	×	B		×		L	0	1
2025	USENIX	Jiachen	def	Ev	DL	Img	✓	✓	✓	B		✓	H	H	0	2
2025	USENIX	Lingchen	def	Ev	DL	Txt	✓	✓	×	W	F	×	L	L	1	2
2025	USENIX	Revisiting	atk	Po	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2025	USENIX	Zhisheng	def	Pr	DL	Aud	✓	✓	×	B		×	L	L	0	1
2025	USENIX	with	def	Ev	DL	Oth	✓	✓	×	W	F	×	H	L	1	3
2025	USENIX	Persistent	atk	Po	DL	Img	✓	✓	×	W		✓	H	L	1	4
2025	USENIX	Benign	atk	Po	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2025	USENIX	34th	def	Ev	DL	Mal	✓	✓	×	B		✓	H	L	0	3